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Original research

Hip adduction and abduction strength and adduction-to-abduction ratio changes across an Australian Football League season

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ABSTRACT

Objectives: Pre-season hip strength testing only represents the athlete's level of conditioning at that time point, and may change over an Australian Football (AF) season. This study aimed to examine if there are changes in hip adduction, abduction and the adduction-to-abduction ratio between preferred and non-preferred kicking legs throughout an AF season. The influence of training load and player characteristics was also examined.
Design: Cross-sectional repeated measures.
Methods: 38 uninjured elite AF players were included. Maximal isometric hip adduction and abduction strength were measured at four time points: start of pre-season (T1), end of pre-season (T2), mid-season (T3) and post-season (T4) using a hand held dynamometer with external belt fixation.
Results: Hip adduction strength and hip-adduction-to-abduction ratio were greater in T3 compared to T1 (adduction by 22.71 N, $p < 0.001$, ratio by 0.15 N, $p < 0.001$) and hip adduction and abduction were weaker in T4 compared to T1 (adduction by 18.6 N, $p = 0.004$, abduction by 24.67 N, $p < 0.001$). No differences were found between the preferred and non-preferred leg in adduction ($p = 0.409$) or abduction ($p = 0.602$) strength. There was an interaction between leg and time point for the adduction-to-abduction ratio; at T3 and T4, the ratio of the preferred kicking leg was significantly lower than the non-preferred kicking leg (T3 by 0.14 N, $p = 0.020$, T4 by 0.15 N, $p = 0.019$). Training load was not significantly associated with strength changes.
Conclusions: Hip strength does change over an AF season. Regular in-season hip strength testing should occur to more accurately reflect player condition compared to one pre-season measurement.
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Practical implications

- Hip adduction strength and the adduction-to-abduction ratio increased from the start of the pre-season to mid-season in uninjured AF players. This may have important implications if using pre-season screening scores to predict in-season groin injury.
- The abduction-to abduction ratio of the preferred kicking leg was significantly lower than the non-preferred kicking leg during mid-season and post-season testing. Therefore caution should be used when using the hip strength of the player's uninjured limb as a clinical indicator for return to play following a unilateral groin strain.
- Training load as measured via session-RPE did not influence isometric hip strength scores.

1. Introduction

In Australian Football (AF), groin strains are considered to be the second most common type of muscle strain experienced.[1] It has been consistently reported that hip adductor muscle weakness precedes the onset of groin injuries in a number of team sports[3,4,5] and a hip adduction-to-abduction strength ratio of less than 80% has been reported to be a strong predictor of groin injury.[2] In an attempt to identify players at risk of sustaining a groin injury, such findings have led to a number of sporting codes implementing pre-season hip strength screenings.[7]

Hip strength screening is also used as a method to determine readiness to return to play following injury.[8] It has been reported that re-gaining an abduction-to-abduction ratio of 90–100%, and

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an adduction strength equal to that of the uninjured side are suitable clinical indicators for return to play following a groin strain.[9] However using the strength of the uninjured side as a reference point, is only suitable if hip strength is symmetrical between the legs in uninjured players. Prendergast et al. reported no significant differences in isometric hip adduction or abduction strength, or the adduction-to-abduction ratio, between preferred and non-preferred kicking legs in elite, sub-elite and amateur AF players.[7] In contrast, Thorborg et al. reported that eccentric hip adduction strength was significantly greater in the preferred kicking leg, than in the non-preferred leg in elite soccer players.[10] It has also been demonstrated in semi-professional soccer players that the preferred kicking leg is significantly stronger in isometric hip adduction and abduction, than the non-preferred kicking leg, however the hip adduction-to-abduction ratio was not different between the legs.[11] However, despite the statistically significant difference in isometric strength between the kicking legs, Thorborg et al. concluded that the difference in strength was within the measurement error of the test procedure, and therefore hip adduction and abduction symmetry could be assumed.[11]

The differences in findings in hip strength symmetry between soccer and AF may be owing to the different kicking techniques utilised in the different football codes.[12,13] However, the discrepancies may also be attributed to the differences in the timing of assessment. Thorborg et al. completed their hip strength screening during the mid-season break,[11] whilst Prendergast et al. recorded hip strength profiles during the pre-season.[7] It is possible that hip strength symmetries change throughout the different phases of a season in response to different workloads required for training and competition; this has yet to be determined. It has been suggested that pre-season testing only represents the athlete's level of conditioning at that particular time point, which has implications if using pre-season musculoskeletal screening for predicting in-season injury occurrence[14,15] and return to play criteria. If changes do occur across a season this may indicate the need for more frequent screening to take place over the entire season and change current screening practices.

Therefore, the aim of this study was to establish if there were any significant changes in isometric hip adduction and abduction strengths between preferred and non-preferred kicking legs across an AF season. The association between training load and player characteristics including age, body mass (BM), years of AFL experience, AFL games played prior to 2016 and AFL games played in 2016 on hip strength profiles at each time point throughout the season was also examined.

2. Methods

Thirty-eight elite AF players were recruited from one AFL club, with all players on the club's 2016 playing list included. Player characteristics including: preferred kicking leg, player age, BM, years of AFL experience, AFL games played prior to 2016 and AFL games played in 2016 were extracted from the club's sports science database (Table 1). Ethical approval was received from the university's Human Research Ethics Committee (RDHS-08-16).

Hip adduction and abduction strength of the preferred and non-preferred kicking legs were measured at four time points throughout the season: start of pre-season, November 2015 (T1); end of pre-season, February 2016 (T2); mid-season, June 2016 (T3); and post-season, August 2016 (T4). All testing sessions were conducted after a minimum two-day break, following a game or training session, to allow for consistent measuring conditions to minimise the effect of fatigue on strength measures. Players were excluded from a single time point due to possible aggravation of a current injury, or if they were unavailable at the time of testing, which occurred for T2 and T4 where the participant number was reduced to 37 (one player was injured) and 33 (five players not available) respectively.

Maximal voluntary isometric hip abduction and adduction measurements were obtained using a hand held dynamometer (HHD; PowerTrack II Commander, JTECH Medical, Salt Lake City, Utah) with external belt fixation.[16] The HHD was calibrated before each testing session, using the system calibration. A single physiotherapist performed all of the strength assessments in accordance with the methods described by Thorborg et al.[16,17] Players were instructed to have a warm up trial, for both abduction and adduction, whereby muscle contractions were performed at 50% of maximal voluntary contraction. This was followed by three contractions at 100% of maximal voluntary contraction, all of which were held for five seconds. The highest of the three maximal trials was recorded and used for analysis. The adduction-to-abduction ratio was calculated based on these measures, by dividing the maximal adduction score by the maximal abduction score.[7] After the warm up trial, players were asked if any pain was experienced or symptoms present with the muscle contraction and allowed to continue if symptom free. To ensure consistency, a standardised command was utilised for the athlete and consisted of "go ahead, push, push, push, push and relax" for five seconds.[7] Between trials, a rest period of 30 s was given to the athlete.[16]

Training load data of individual players were recorded throughout the 2016 AF season for each training session and game played. Activities included in the calculation of training load were: on field training sessions, rehabilitation running sessions, weights sessions, game play, lower limb cardiopulmonary sessions and cross training sessions. Training load was calculated based on the player's session rating of perceived exertion (session-RPE),[18] on a ten-point scale to quantify the intensity of the physical activity undertaken.[19] Session-RPE was then multiplied by the activity duration in minutes, resulting in a training load figure in arbitrary units (session-RPE x duration = training load).[18] Total training load was then calculated for the different training periods throughout the season: preseason period, 23/11/15–31/01/16, first half of home/away season, 1/02/16–22/06/16, and second half of the home/away season, 23/06/16–29/08/16, for each participant. Training load was not recorded during the off-season (prior to the beginning of the pre-season), as no formal training was required from the players. Training load was representative of the mean training load calculated for each training period, as there was a different number of training days in each period (pre-season = 70 days; first half of season = 143 days and second half of season = 68 days).

Data were analysed using Stata/IC 15.0 (StataCorp. 2015. *Stata Statistical Software: Release 14*. College Station, TX: StataCorp LP). Descriptive statistics were completed for player demographic characteristics at each time point throughout the season and are described as means and standard deviations. Adjusted linear mixed regression models, with player characteristics included as covariates (training load, age, BM, years of AFL experience, AFL games played prior to 2016 and AFL games played in 2016) were used to examine changes in isometric hip strength outcomes for the preferred and non-preferred kicking legs across the four time points throughout the season. A separate regression model was fitted for each hip strength measure (adduction, abduction and adduction-to-abduction ratio). Prior to multivariable analysis, univariate regression analyses were conducted to identify which of the player characteristics would potentially contribute to the final model. The final multivariable model included only variables shown to significantly influence the hip strength measure. Prior to interpreting the results of the models, several assumptions were evaluated, confirming that each continuous variable was approximately normally distributed. The results from the regression analyses were reported

Table 1
Player demographics at each time point throughout the season. Data are presented as means (SD) [range].

Player characteristic	All *n* = 38	T1 *n* = 38	T2 *n* = 37	T3 *n* = 38	T4 *n* = 33
Average training load	NA	0	188.2 (30.6) [103.3–248.1]	200.0 (16.7) [153.9–231.9]	184.6 (35.0) [25.0–251.0]
Age (years)	23.0 (3.5) [18–30]	23.0 (3.5) [18–30]	23.2 (3.4) [18–30]	23.0 (3.5) [18–30]	22.8 (3.6) [18–30]
Body mass (kg)	87.1 (7.5) [73.7–102.1]	87.1 (7.5) [73.–102.1]	87.3 (7.5) [73.7–102.1]	87.1 (7.5) [73.7–102.1]	86.9 (7.7) [73.7–102.1]
AFL years' experience	5.9 (3.6) [1–13]	5.9 (3.6) [1–13]	6.0 (3.5) [1–13]	5.9 (3.6) [1–13]	5.8 (3.6) [1–13]
AFL games prior 2016	61.7 (59.2) [0–197]	61.7 (59.2) [0–197]	63.4 (59.1) [0–197]	61.7 (59.2) [0–197]	61.9 (60.9) [0–197]
AFL games in 2016	8.4 (7.9) [0–19]	8.4 (7.9) [0–19]	8.65 (7.9) [0–19]	8.4 (7.9) [0–19]	8.5 (7.8) [0–19]

AFL = Australian Football League, T1 = start of pre-season, T2 = end of pre-season, T3 = mid-season, T4 = post-season.

Table 2
Means (SD) for isometric hip adduction and abduction strength (newtons; N) and the adduction-to-abduction ratio for the preferred and non-preferred kicking legs at four time points throughout the season.

Hip Strength Measure	Leg	T1 *n* = 38	T2 *n* = 37	T3 *n* = 38	T4 *n* = 33
Adduction (N)	Preferred	186.9 (35.0)	183.6 (45.5)	203.8 (48.3)	158.9 (39.2)
	Non-preferred	182.3 (38.6)	183.3 (39.7)	203.7 (37.9)	158.9 (39.2)
Abduction (N)	Preferred	171.2 (34.0)	171.0 (34.1)	183.5 (32.1)	156.9 (31.9)
	Non-preferred	172.4 (32.0)	165.9 (31.3)	174.1 (38.2)	148.0 (30.6)
Adduction-to-abduction ratio	Preferred	1.11 (0.18)	1.09 (0.26)	1.11 (0.17)	1.01 (0.15)
	Non-preferred	1.08 (0.23)	1.12 (0.20)	1.21 (0.31)	1.11 (0.20)

T1 = start of pre-season, T2 = end of pre-season, T3 = mid-season, T4 = post-season.

as regression coefficients, with 95% confidence intervals. For all analyses a p value of less than 0.05 was considered statistically significant.

3. Results

The means and standard deviations of isometric hip adduction and abduction strength and the adduction-to-abduction ratio, for the preferred and non-preferred kicking legs are presented in Table 2.

The univariate models demonstrated that training load, player age, years of AFL experience, AFL games played prior to 2016 and AFL games played in 2016 were not associated with isometric hip adduction or hip abduction strength or the adduction-to-abduction ratio. Body mass was not associated with hip adduction strength, but did show evidence of an association with isometric hip abduction strength and the adduction-to-abduction ratio. For every one kg increase in BM, there was a 0.989 N increase in hip abduction strength ($p = 0.008$, 95% CI [0.262–1.715]), and a 0.007 decrease in the adduction-to-abduction ratio ($p = 0.008$, 95% CI [−0.011 to 0.012]).

The final multivariable model for isometric hip adduction strength demonstrated that in comparison to T1, the hip adductors were significantly stronger overall (by 22.71 N, $p < 0.001$) at T3, and significantly weaker (by 18.6 N, $p = 0.004$) at T4 (Table 3). No differences were observed in hip adduction strength between T1 and T2 ($p = 0.860$). There was no difference in hip adduction strength between the preferred and non-preferred kicking legs ($p = 0.409$). No interaction between leg and time point for hip adduction strength was evident (Table 3).

The final multivariable model for isometric hip abduction strength demonstrated that when controlling for BM, in comparison to T1, the hip abductors were significantly weaker at T4 (by 24.67 N, $p < 0.001$) (Table 3). There was no difference in hip abduction strength between T1 and T2 ($p = 0.098$) or T1 and T3 ($p = 0.842$). No difference was observed in hip abduction strength between the preferred and non-preferred kicking legs ($p = 0.602$) nor was there an interaction between leg and time point for hip adduction strength (Table 3).

The final multivariable model for the hip adduction-to-abduction ratio demonstrated that when controlling for BM, in comparison to T1, the ratio was significantly greater at T3 (by 0.15 N, $p < 0.001$) (Table 3). Hip adduction-to-abduction ratio between time points T1 and T2 ($p = 0.354$) and T1 and T4 ($p = 0.140$) resulted in no differences. There was no difference in the hip adduction-to-abduction ratio between the preferred and non-preferred kicking legs ($p = 0.329$). An interaction between leg and time point for the hip adduction-to-abduction ratio (Table 3) was evident whereby at T3 and T4, the abduction-to abduction ratio of the preferred kicking leg was 0.14 N ($p = 0.020$) and 0.15 N ($p = 0.019$) lower than the non-preferred kicking leg.

4. Discussion

This is the first study to measure isometric hip adduction and abduction strength profiles of uninjured elite AF players at different time points throughout an AF season. The results demonstrate that isometric hip adduction strength, hip abduction strength and the adduction-to-abduction ratio do change throughout an AF season. The symmetry between preferred and non-preferred kicking legs varied at different time points throughout the season and training load was not associated with hip strength changes.

There were no significant differences in any of the hip strength measures between the start of the pre-season (T1) and the end of the pre-season (T2). This indicates that a single assessment during the pre-season at a time that is convenient for the club is sufficient. This could reduce the burden of repeated measures on players and staff.

The significant increase in hip adduction strength and the adduction-to-abduction ratio from the start of the pre-season to mid-season in uninjured AF players is consistent with the findings of Wollin et al. who measured hip strength during the pre-season and then monthly during the in-season in elite male youth soccer players. They demonstrated that hip adductor strength and the adduction-to-abduction ratio were lowest during the pre-season and significantly increased by month two, with further increases seen as the season progressed.[20] Additionally, Hides et al. measured the size of key trunk muscles over three AFL seasons via an MRI.[21] They found the size of torque producing muscles (lumbar erector spinae and internal oblique) increased over the playing season and decreased by the start of the next season. Both Wollin et al. and Hides et al. suggested that the increases in strength and

Table 3
Final multivariable models for changes in isometric hip strength measurements at four time points throughout an Australian Football League season.

Hip strength measure	Covariate	Coefficient	p Value	95% CI	
				Lower	Upper
Adduction	T2	−1.07	0.860	−12.92	10.78
	T3	22.71	<0.001*	10.48	34.94
	T4	−18.60	0.004*	−31.37	−5.84
	Preferred Leg	4.89	0.409	−6.71	16.49
	T2#Preferred leg	−2.00	0.814	−18.69	14.68
	T3#Preferred leg	−4.25	0.627	−21.37	12.88
	T4#Preferred leg	−4.89	0.590	−22.68	12.90
Abduction	T2	−8.76	0.098	−19.14	1.62
	T3	1.09	0.842	−9.61	11.78
	T4	−24.67	<0.001*	−35.83	13.51
	Preferred leg	−2.74	0.602	−13.05	7.56
	T2#Prefered leg	6.93	0.355	−7.77	21.62
	T3#Preferred leg	12.38	0.107	−2.69	27.46
	T4#Preferred leg	12.68	0.112	−2.97	28.34
	Body mass (kg)	0.99	0.008*	0.26	1.72
	T2	0.04	0.354	−0.04	0.12
Adduction-to-abduction ratio	T3	0.15	<0.001*	0.07	0.23
	T4	0.07	0.140	−0.02	0.15
	Preferred leg	0.04	0.329	−0.04	0.12
	T2#Preferred leg	−0.04	0.463	−0.16	0.07
	T3#Preferred leg	−0.14	0.020*	−0.26	−0.02
	T4#Preferred leg	−0.15	0.019*	−0.27	−0.02
	Body mass (kg)	−0.01	0.008*	−0.01	−0.002

T1 = start of pre-season, T2 = end of pre-season, T3 = mid-season, T4 = post-season, CI = confidence interval.
* $p < 0.05$.

muscle size were owing to larger workloads and training demands during the in-season, however neither study measured training load.[20,21] Nevertheless the fact that hip strength increases as the season progresses has significant implications for the use of pre-season musculoskeletal screening for in-season injury prediction. Performing a test during the pre-season only tells you about the athletes' condition at that particular time point, which will vary through a season with exposure to training and competition.[15] More frequent in-season testing to track a player's strength and individual response to changing workloads throughout the season is required, before hip strength can be used as a method to predict injuries.

There was a significant decrease in both hip adduction and abduction strength between start of pre-season and the post-season in uninjured AF players. Testing players during the off-season is challenging due to player availability and anecdotally player effort during post-season testing is reduced which may have influenced the results. Hip strength testing during the post-season is unlikely to assist with the prediction of injuries during the season due to the limitations and timing of testing. Instead, additional testing time points throughout the in-season are likely to provide more reliable information about player hip strength changes.

The results of this study demonstrate that there is symmetry in isometric hip adduction and abduction strength between the preferred and non-preferred kicking legs during the pre-season. This is in line with the results of Prendergast et al. who demonstrated that there is hip strength symmetry between preferred and non-preferred kicking legs in elite, sub-elite and amateur AF players when measured during the pre-season.[7] However there was an interaction between kicking leg and time point for the adduction-to-abduction ratio. At the mid-season and post-season time points, the abduction-to-abduction ratio of the preferred kicking leg was significantly lower than that of the non-preferred kicking leg, which appears to be the result of greater abduction strength in the preferred kicking leg at these time points. This is consistent with previous research that has demonstrated that muscle size asymmetries exist between the preferred and non-preferred kicking leg in AF players.[22,23] This may be due to the fact that AF involves many asymmetrical skills and footballers rarely utilise their lower limbs with equal preference, selectively using the dominant or preferred leg for most game based activities such as kicking, changing direction and jumping.[24,25] During the pre-season players are likely to train both legs equally as it is desirable to kick competently with both limbs to gain a tactical advantage,[24] however during competition footballers tend to develop a preferred kicking leg and use their dominant movement patterns to achieve optimal outcomes in pressure situations.[22,23] Game analysis of variables such as kicking prevalence is required to quantify the use of the preferred and non-preferred limbs in conjunction with in-season hip strength testing. This may assist in identifying changes in individual loading patterns which may provide better injury prediction criteria than pre-season screening alone, however this requires further investigation. Nevertheless, coaching and sports science staff should use caution when using the hip strength of the player's uninjured limb as a clinical indicator for return to play following a unilateral groin strain, as it appears an imbalance occurs throughout the season, even in uninjured players.

Despite finding changes in hip strength scores throughout the season in uninjured AF players, this was not associated with training load. This is consistent with the findings of Esmaeili et al. who found that various cumulative and relative measures of training load as measured via session-RPE, had no effect on weekly musculoskeletal screening scores (sit and reach test, dorsiflexion lunge test and adductor squeeze test) in elite AF players, indicating that screening scores did not appear to be affected by regular exposure to training and competition.[26] They proposed that technique variation, equipment error, or a true change in an athletes' performance (e.g. adaptation to training and competition, residual effects of complete resolution from previous injuries) may contribute to the week-to-week changes in test scores.[26] It is also worth considering whether session-RPE is sensitive enough to measure changes in individual player load. The proportion of training spent performing sports-specific skills (e.g. kicking) compared with general strength and conditioning may need to be considered, as the intensity and volume of each type of training changes during different phases of the season. Additionally, player statistics obtained from match

analysis and global positioning system variables such as number of kicks, disposals, tackles, distance covered, and speed, may provide more quantitative data on individual workload, which may have a better association with hip strength, however this requires further investigation. More frequent in-season testing to track a player's strength and individual response to changing loads throughout the season is required, to provide a more in depth understanding of potential injury risk, which requires further investigation.

5. Conclusion

This study established that there was a significant increase in hip adduction strength and the adduction-to-abduction ratio from the start of the pre-season to mid-season in uninjured AF players. This change in hip strength across an AF season as well as differences between the preferred and non-preferred leg may have significant implications if using one off pre-season hip strength measurements as an indicator of increased risk of groin injury and return to play. We propose that regular in-season testing be completed at various time points throughout the season to ensure these changes in strength better reflect player condition compared to one pre-season measurement.

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